

Serie 19

EIGENVECTORS, EIGENVALUES

1. In each of the following cases, let T_i be the endomorphism of $\mathbb{R}^2$ which is represented by the matrix A_i in the standard ordered basis for $\mathbb{R}^2$, and let U_i be the endomorphism of $\mathbb{C}^2$ represented by A_i in the standard ordered basis. Find the characteristic polynomial for T_i and that for U_i , find the eigenvalues of each endomorphism, and for each such eigenvalue find a basis for the corresponding space of eigenvectors.

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

2. Let K be a field and let V be a finite-dimensional vector space over K . Suppose that $T \in \text{End}(V)$ is invertible. Prove that $\text{Eig}_T(\lambda) = \text{Eig}_{T^{-1}}(1/\lambda)$ for every $\lambda \in K^*$.
3. Consider the space $C^\infty(\mathbb{R})$ of smooth functions over $\mathbb{R}$ and the map

$$\begin{array}{ccc} T : C^\infty(\mathbb{R}) & \rightarrow & C^\infty(\mathbb{R}) \\ f & \mapsto & f' \end{array}$$

Find the eigenvalues and the corresponding eigenfunctions (this is a synonym for eigenvectors when working on a space whose elements are functions) of T .

4. Let $K = \mathbb{R}$, show that K^∞ does not admit any countable basis.
Hint: Use the fact that pairwise distinct eigenvalues correspond to a set of linearly independent eigenvectors.
5. Let K be a field and let V be an n -dimensional vector space over K ($n > 0$).
 - (a) Let T be a diagonalizable endomorphism of V with (not necessarily distinct) eigenvalues λ_i for $1 \leq i \leq n$. Show that

$$\text{Tr}(T) = \sum_{i=1}^n \lambda_i \quad \text{and that} \quad \det(T) = \prod_{i=1}^n \lambda_i.$$

For $0 \leq k \leq n$, let c_k be the coefficient of x^k in the characteristic polynomial of T . Give a formula for c_k in terms of the eigenvalues of T .

- (b) Let $B \in M_{2 \times 2}(\mathbb{R})$ be diagonalizable with $\text{Tr}(B) = 0$. Show that $\det(B) \leq 0$.